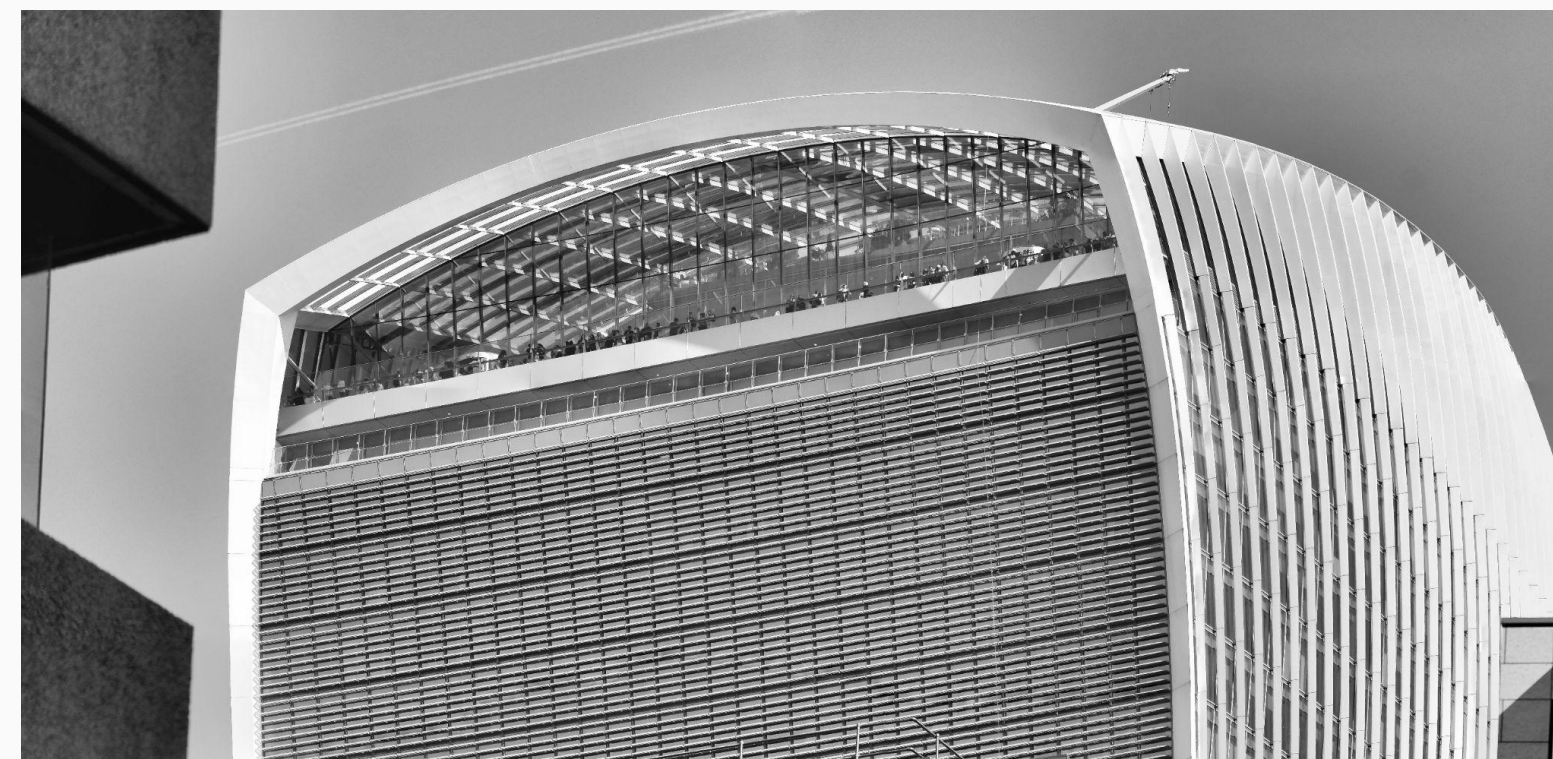


KESTREL BANK

Business Loan Risk Model

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BUSINESS TASK

Kestrel Bank wants to flag financial distress in borrowers earlier. Banks usually act only after a missed payment — by then, the company may already be insolvent.

Public company records show warning signs sooner:
officer resignations, late filings, new charges.



Scope: Ltd, PLC & LLP companies Kestrel
Bank's credit risk team

HYPOTHESIS

Companies that go on to fail show these signals more often, and earlier, than companies that stay active.

Overdue Filings

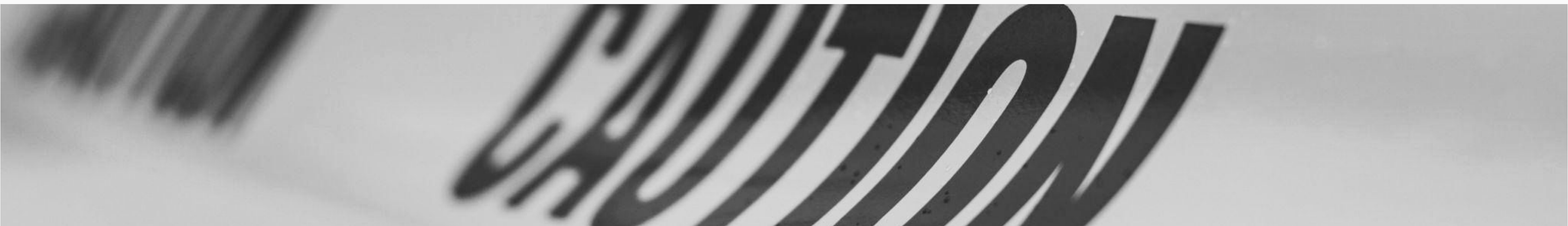
A required filing becomes late — an early compliance warning sign.

Officer Resignations

Directors and company secretaries start leaving before things go wrong.

New Charges

A new claim is registered against the company's assets.





DATA SOURCES

1. Free Company Data

Bulk download, all 5.5m registered UK companies, refreshed monthly.

2. Officer Records

Director & secretary appointments and resignations, pulled per company via the API.

3. Filing History

Insolvency filings — a real, dated failure event.

4. Registered Charges

Claims against company assets, usually securing a loan.

5.5m UK companies → 113,679 in formal insolvency → 10,000 balanced sample → 9,844 final (Ltd/PLC/LLP only)

BUILDING THE RISK SCORE

Three signals, tracked over 12 months, combine into a single score:
Director resignation + Overdue filing + New charge = Risk Score (0–3).

12-Month Signal (Avg.)	Active Companies	Failed Companies
Director resignations	0.13	0.29
Overdue at reference date	5.9%	47.4%
Avg. total charges	0.64	1.35
Avg. outstanding charges	0.33	0.74

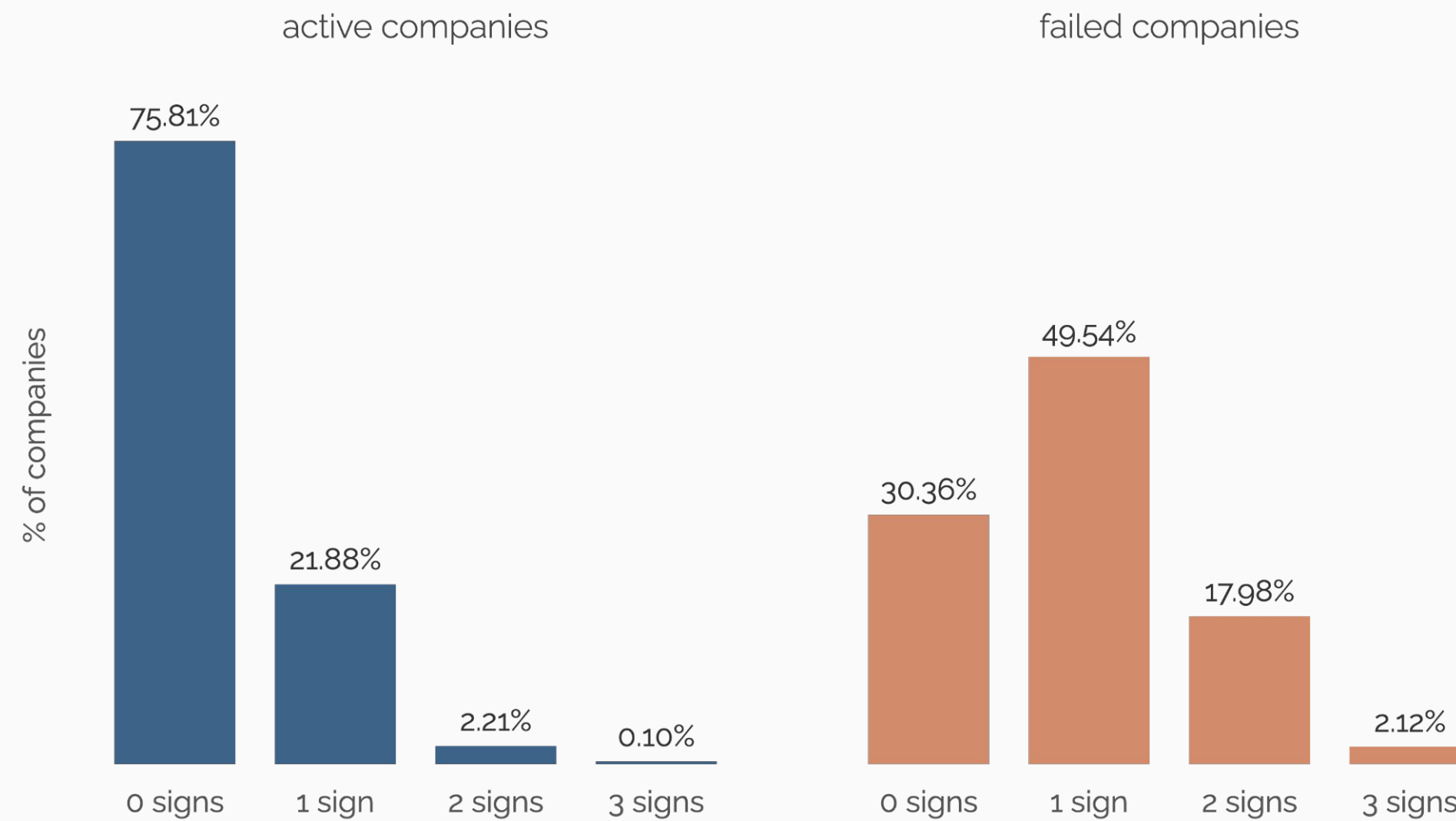
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~8x

Failed companies are about 8 times more likely to have an overdue filing

5.9% of active companies, vs. 47.4% of failed companies

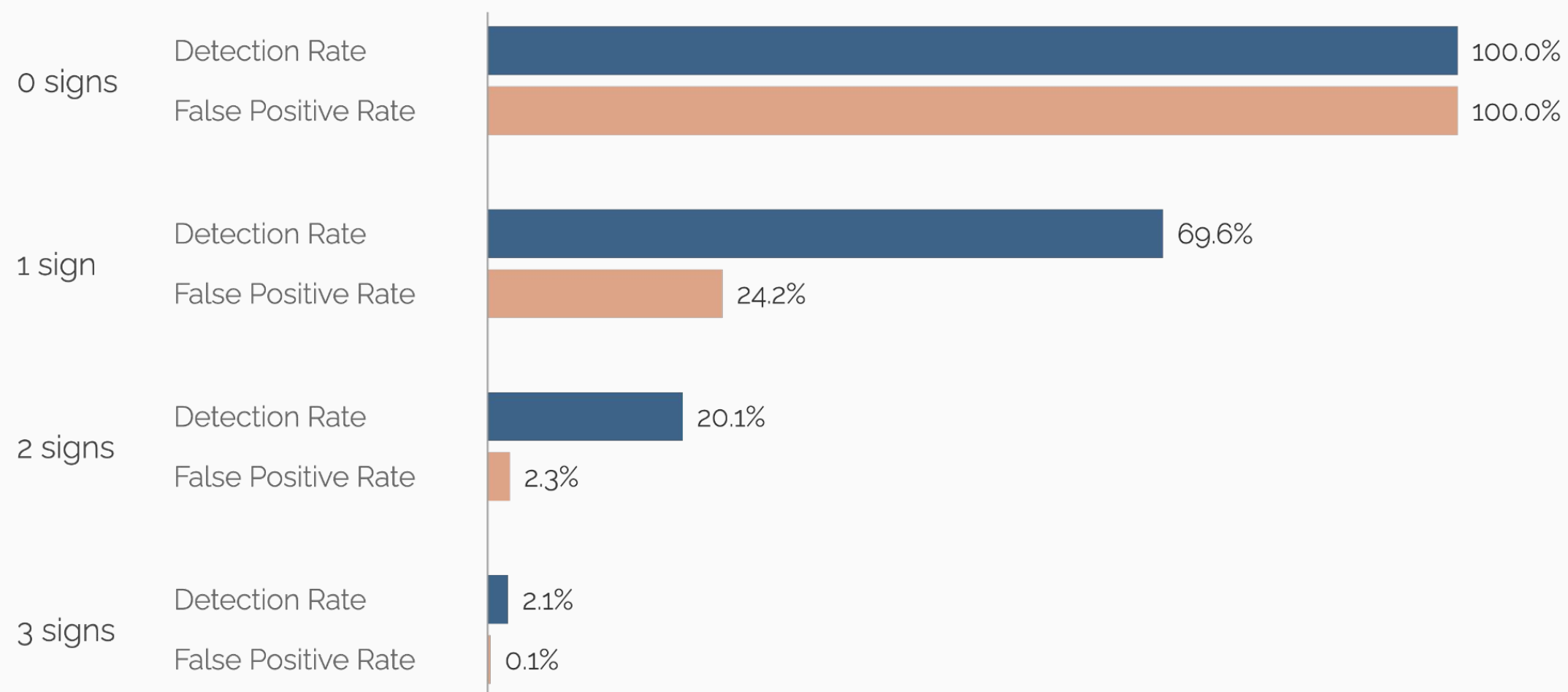
RISK SCORE DISTRIBUTION



According to the model:

- Most active companies show no warning signs at all
- Failed companies are far more likely to show at least one
- The gap between the two groups widens with every additional sign

RISK MODEL PERFORMANCE



- 1 sign — Model catches most failures, but also flags a lot of healthy companies
- 2 signs — Model sharply cuts false positives while still catching 1 in 5 failures
- 3 signs — Model is nearly noise-free, but too narrow to rely on alone

LIMITATIONS

what this model doesn't account for...



Case-control sample

Scores show relative risk, not real-world probability

Archival gaps

16% of failed companies predate digital records

Scope

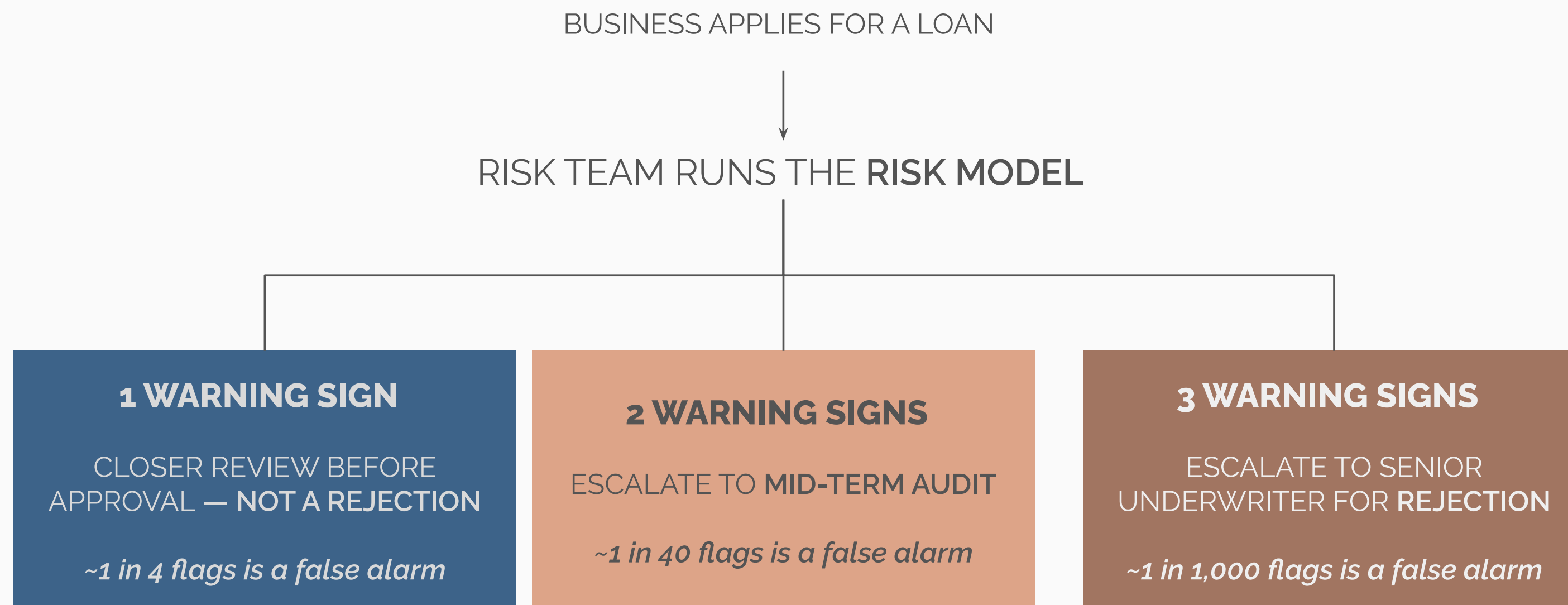
Sole traders and non-corporate entities sit outside the model

No fixed threshold

The right cut off depends on how Kestrel Bank uses it

RECOMMENDATIONS

PUTTING THE MODEL TO WORK...



* The model informs credit decisions — it doesn't automate them.
These flag points are a starting proposal for Kestrel Bank's credit team to test and adjust.



THANK YOU!

HAPPY TO ANSWER ANY QUESTIONS



Emanuel Cunha



Data Analyst



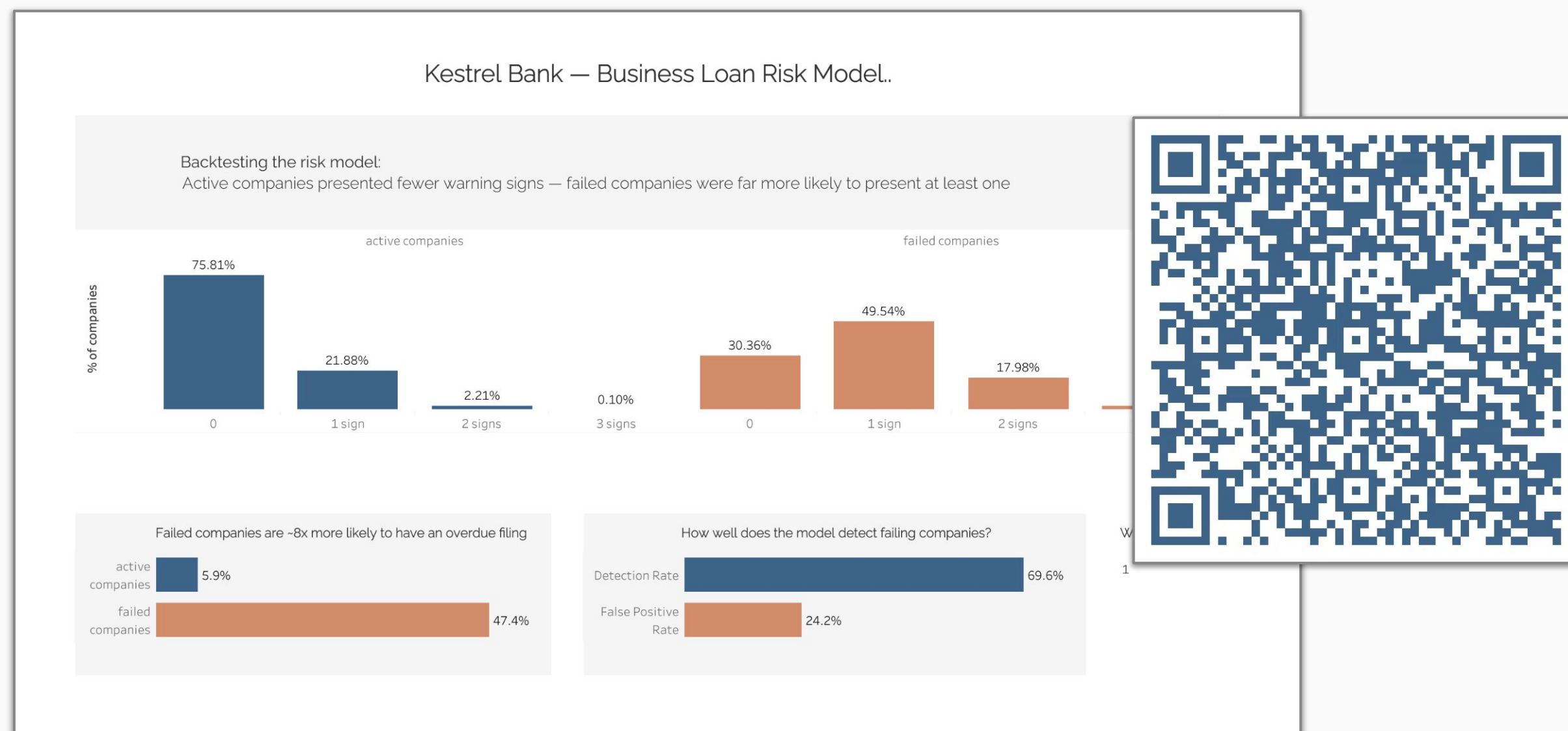
Emanuel.cunha.uk@gmail.com



www.linkedin.com/in/emanuel-cunha-uk

EXPLORE LIVE

THE FULL ANALYSIS INTERACTIVE



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